CSCI0220 Midterm Solutions

Due: *June 29th at 11:59pm EDT*

Please read in full:

You are not allowed to use the internet outside of the CSCI0220 website. Doing so is a breach of the **Collaboration Policy** and **Brown's Academic Code**.

You can refer to lecture slides, lecture captures, your own notes, the course textbook, the website resources, and Campuswire. You must complete the entire exam by yourself, without discussion of any kind with anyone else.

By submitting this exam, you are confirming that you have understood these policies, and that you have followed them.

As always, please do not include any identifying information about yourself in the handin, including your Banner ID. Be sure to fully explain your reasoning and show all work for full credit.

Problem 1

a. Prove using the set-element method that for finite sets X and Y with the same universal set U

$$X \cap Y = (X \cup Y) - (\overline{X} \cup \overline{Y}).$$

b. i. Consider an arbitrary binary operation $\star$ on a finite set X. A *left identity element* for $\star$ on the set X is any $e \in X$ such that $e \star x = x$ for all $x \in X$. For example, 1 is the left identity element for the multiplication operator '$\times$' on the integers $\mathbb{Z}$ because $1 \times x = x$ for all $x \in \mathbb{Z}$.

Find the left identity elements of the set difference operator '$-$' on the set $\mathcal{P}(S)$, where S is an arbitrary non-empty, finite set. If there exists one, **prove** why the element you chose is an identity element, **and prove** why no other element would be. If there does not exist one, **prove** that no element can be.

ii. By the same logic, a *right identity element* for $\star$ on the set X is any $e \in X$ such that $x \star e = x$ for all $x \in X$. For example, 1 is the right identity element for the division operator '/' on the integers $\mathbb{Z}$ because $x/1 = x$ for all $x \in \mathbb{Z}$.

Find the right identity elements of the set difference operator '$-$' on the same set $\mathcal{P}(S)$. If there exists one, **prove** why the element you chose is an identity element, **and prove** why no other element would be. If there does not exist one, **prove** that no element can be.

Solution

a. We use the element method in both directions.

1. Suppose $x \in X \cap Y$. By definition of set difference, $x \in (X \cup Y) - (\overline{X} \cup \overline{Y})$ if $x \in (X \cup Y)$ AND $x \notin (\overline{X} \cup \overline{Y})$. Since $x \in X \cap Y$, we know that $x \in X$ and $x \in Y$. Therefore, by definition of set union, $x \in X \cup Y$, which means that $x \notin \overline{X}$ and $x \notin \overline{Y}$, or $x \notin \overline{X} \cup \overline{Y}$. Thus, $x \in (X \cup Y) - (\overline{X} \cup \overline{Y})$ and we conclude that $X \cap Y$ is a subset of $(X \cup Y) - (\overline{X} \cup \overline{Y})$.
2. Suppose $x \in (X \cup Y) - (\overline{X} \cup \overline{Y})$. Then, $x \in X \cup Y$ and $x \notin (\overline{X} \cup \overline{Y})$. Since $x \notin (\overline{X} \cup \overline{Y})$, then x must be in the complement of $(\overline{X} \cup \overline{Y})$. Thus, by De Morgan's law, $x \in X \cap Y$. Thus, $(X \cup Y) - (\overline{X} \cap \overline{Y})$ is a subset of $X \cap Y$.

We have shown both directions. Thus, by the set-element method, we conclude that $X \cap Y = (X \cup Y) - (\overline{X} \cup \overline{Y})$.

b. i. For the set difference operator, there does not exist a left identity element on $\mathcal{P}(S)$. Let us prove this statement.

Proof. Let us assume for the sake of contradiction that there exists a **non-empty** left identity for the set difference operator, which we'll denote by e where $e \in \mathcal{P}(S)$. Therefore, for all $x \in \mathcal{P}(S)$, we know that $e - x = x$. We know by definition of the set difference that $e - x$ consists of every element in the subset e that is not in the subset x. Therefore, $\forall y \in e - x$, $y \notin x$. Thus, $e - x \neq x$, which is a contradiction. We conclude that there does not exist a non-empty left identity for the set-difference operator.

We now consider the case where e is **empty** such that $e = \{\}$. By definition of the set difference $e - x = \{\}$. Thus, if $|x| > 0$, $e - x \neq x$ and we conclude that $e = \{\}$ is not a left identity either.

We have shown that no element $e \in \mathcal{P}(S)$ works as an left identity element. Thus, there does not exist a left identity element on $\mathcal{P}(S)$ for the set-difference operator. $\square$

ii. For the set-difference operator, the right identity element is $r = \{\}$. Let us prove this statement.

Proof. By definition of the set difference, $x - r$ consists of every element in the subset x that is not in the subset r. By definition, r consists of no elements. Hence, x and r have no elements in common. Therefore, the set difference consists of exactly every element in x such that $x - r = x$. $\square$

We now demonstrate that no other element is the right identity element. Consider a non-empty $o \in \mathcal{P}(S)$. Let us demonstrate that o is not a right identity element.

Proof. Assume for the sake of contradiction that o is a right identity element. Therefore, for all $x \in \mathcal{P}(S)$, we have $x - o = x$. Since o is non-empty, it consists of at least one element $y \in S$. We now consider an $x \in \mathcal{P}(S)$ such that $y \in x$. Since $y \in o$ and $y \in x$, we find that $y \notin x - o$. Thus, there exists an x such that $x - o \neq x$, which is a contradiction. We conclude that o is not a right identity element. $\square$

Problem 2

Let B be the set of all possible propositional formulas. Note that B is a set of infinite cardinality. Define the relation R on B such that, for $a, b \in B$, $a \; R \; b$ if and only if a and b always return the same output given the same input.

a. Let x be a proposition with Boolean inputs p, q and r and the following truth table:

p	q	r	x
1	1	1	1
1	1	0	0
1	0	1	0
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	1
0	0	0	0

Write a propositional formula corresponding to x in conjunctive normal form with a maximum of two clauses.
Note: Explain how you arrived at your answer.

b. Prove that R is an equivalence relation.

c. Create a bijection between the equivalence classes of three-input, one-output propositional formulas and binary strings of length eight.
Note: You need not prove your answer. You must just explain the reasoning behind your bijection.

d. How many equivalence classes of three-input, one-output propositional formulas are there?
Note: You must thoroughly explain your answer, but are not required to prove it.

e. The equivalence class of a propositional formula x is $[x]_R = \{a \in B \mid (a, x) \in R\}$. Prove that $|[x]_R|$ is infinite for any $x \in B$.

Solution

a. Here is a correct example:

r AND (NOT(p) OR q)
This propositional formula can be found by noticing that r is True in all three True outputs of x, and that the combinations of p and q that make x output True when r is True is when $p = 1$ and $q = 1$, when $p = 0$ and $q = 1$, and when $p = 0$ and $q = 0$. Hence, the term (NOT(p) OR q) covers these cases.

b. To show that R is an equivalence relation, we shall show that it is reflexive, symmetric, and transitive.

Reflexive

Consider an arbitrary propositional formula a. Since a returns the same input as itself on any given input, aRa. Thus, R is reflexive.

Symmetric

Consider arbitrary propositional formulas a and b such that $a\ R\ b$. Since a and b both return the same output on any given input, we know that $b\ R\ a$. Thus, R is symmetric.

Transitive

Consider arbitrary propositional formulas a, b, and c such that $a\ R\ b$ and $b\ R\ c$. Since b returns the same output as a on any given input, and c returns the same output as b on any given input, we may conclude that a and c also return the same output on any given input such that $a\ R\ c$. Thus, R is transitive.

c. Each equivalence class consists of propositional formulas whose outputs are the same given the same inputs. Considering the three-input, one-output propositional formulas, since there are three inputs, each of which can be true or false, there are $2^3 = 8$ possible input combinations. This corresponds to eight rows of the truth table of any propositional formula in these equivalence classes.

Each propositional formula in an equivalence class has the same output for each of the eight rows. Numbering the eight rows of distinct input combinations from one to eight, we create a length eight binary string by putting the output of row $1 \leq n \leq 8$ in digit position n from the right of the length eight binary string.

Since each equivalence class has a corresponding truth table output combination, every equivalence class can be mapped in this way to a distinct length eight binary string.

d. Recall that for two propositional formulas to be in distinct equivalence classes, there must be at least one set of inputs for which the expressions produce a different output. Therefore, to count the number of equivalence classes, we can count the number of different ways to assign outputs to each set of inputs.

Let us begin this problem by counting the number of possible combinations of inputs to the expressions. Since there are three inputs, each of which can be true or false, there are $2^3 = 8$ possible input combinations.

Let us now consider the possible outputs of these binary functions. For each of the 8 input sets, a binary expression can return either True or False. Therefore, there are 2^8 different "output possibilities," or ways that expressions can differ in outputs when given the same set of possible inputs. Thus, there are $2^8 = 256$ distinct equivalence classes.

e. Claim: $|[x]_R|$ is infinite.

Proof. Assume, for the sake of contradiction, that there is some $x \in B$ for which $|[x]_R|$ is equal to some finite integer k.

If $[x]_R$ is finite, then it contains some proposition m with a number of logical operators that is greater than or equal to all the other propositions in $[x]_R$. m is, in a sense, the 'largest' element of $[x]_R$

We can create a distinct Boolean expression by for example ANDing tautologies or ORing contradictions at the end of m, or by putting m in some even number of consecutive NOTs without any modification to the Boolean output with respect to the original propositional formula m.

This new propositional formula will produce exactly the same output as the original propositional formula m on any given input, and so it will also be in $[x]_R$. However, we took m to be the 'largest' element of $[x]_R$: We reached a contradiction.

Since we have contradicted our assumption that a finite equivalence class can exist, we conclude that $|[x]_R|$ is infinite for any $x \in B$. □

Problem 3

Let the relation $R_{15} = \{(a, b) \mid \exists i \in \mathbb{Z}$ such that $(a - b) = 15i\}$. Prove by induction that, for any finite set S with more than one element, $(|\mathcal{P}(\mathcal{P}(S))|, 1) \in R_{15}$.

Solution

Proof. We prove this proposition by induction

Proposition:

The proposition $P(n)$ we wish to prove is "For any finite set S with cardinality $n \geq 2$, $(|\mathcal{P}(\mathcal{P}(S))|, 1) \in R_{15}$."

Thus, by the definition of the relation R_{15}, we want to show that the difference between $|\mathcal{P}(\mathcal{P}(S))|$ and 1 is some multiple of 15 such that for some $i \in \mathbb{Z}$

$$|\mathcal{P}(\mathcal{P}(S))| - 1 = 15i$$

Base Case: Consider the case $n = 2$.

Here we consider a set S_2 such that $|S_2| = 2$. We find that

$$\begin{aligned}
|\mathcal{P}(\mathcal{P}(S_2))| - 1 &= 2^{|\mathcal{P}(S_2)|} - 1 \\
&= 2^{(2^{|S_2|})} - 1 \\
&= 2^{(2^2)} - 1 \\
&= 2^4 - 1 \\
&= 16 - 1 \\
&= 15
\end{aligned}$$

Thus, if we let $i = 1$, we see that $|\mathcal{P}(\mathcal{P}(S_2))| - 1 = 15i$. Therefore, the base case $P(2)$ is true.

Inductive Hypothesis: Assume $P(k)$ is true for $n = k$.

The inductive assumption is that for some set S_k such that $|S_k| = k$, there exists some integer value i for which

$$|\mathcal{P}(\mathcal{P}(S_k))| - 1 = 15i$$

Since the powerset of a set of cardinality k has the cardinality 2^k, we find that $|\mathcal{P}(\mathcal{P}(S_k))| = 2^{(2^k)}$ such that our assumption leads to $2^{(2^k)} - 1 = 15i$ for some $i \in \mathbb{Z}$.

Inductive Step: Using the inductive hypothesis, we now show that $P(k+1)$ is true.

The inductive step is to show that for a set S_{k+1} such that $|S_{k+1}| = k+1$, there exists some integer value of j for which $|\mathcal{P}(\mathcal{P}(S_{k+1}))| - 1 = 15j$ holds.

We put to use our inductive assumption that $2^{(2^k)} = 15i + 1$ for some $i \in \mathbb{Z}$:

$$\begin{aligned}
|\mathcal{P}(\mathcal{P}(S_{k+1}))| - 1 &= 2^{|\mathcal{P}(S_{k+1})|} - 1 \\
&= 2^{(2^{n+1})} - 1 \\
&= 2^{(2 \cdot 2^n)} - 1 \\
&= (2^{(2^n)})^2 - 1 \\
&= (15i + 1)^2 - 1 \\
&= (225i^2 + 30i + 1) - 1 \\
&= 15(15i^2 + 2i) + 1 - 1 \\
&= 15(15i^2 + 2i)
\end{aligned}$$

Thus, if we let $j = 15i^2 + 2i$, then we see that $|\mathcal{P}(\mathcal{P}(S_{k+1}))| - 1 = 15j$. We conclude that if $P(k)$ is true, then $P(k+1)$ is true.

Conclusion:

We have shown that $P(2)$ is true. We have also shown that if $P(k)$ is true, then $P(k+1)$ is true. Hence, we have demonstrated by induction that $P(n)$ is true for all $n \geq 2$. In other words, for any finite set S with more than one element, $(|\mathcal{P}(\mathcal{P}(S))|, 1) \in R_{15}$.

$\square$